

Lecture 11-13

→ Backbone of Statistical Mechanics, (classical form of QFT (WKB), stochastic physics ...)

↳ We set out to make predictions about turbulent and chaotic systems

- nonlinear systems being used
1. Count (possible configurations)
 2. Weight (based on occupation size in state space)
 3. Predict
- Linearize: trade 3 dimensions for ∞ -many dimensions

Transporting density (of initial conditions)

→ in state space, we only ever have small volumes.

→ There are never points in state space, since nothing is measured with infinite precision

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19.1

measures

(not reality)

(reality)

course graining



← n-dimensional Gaussian/normal dist. is our ensemble of initial conditions

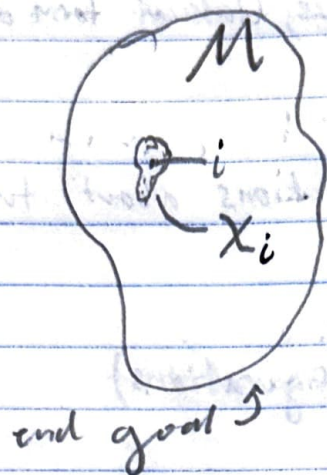
Measure Theory

$$d\mu(x) = p(x) dx \quad (p(x) = p(x, t))$$

→ Linear stability analysis tells us that the "cloud" of initial conditions spread out linearly

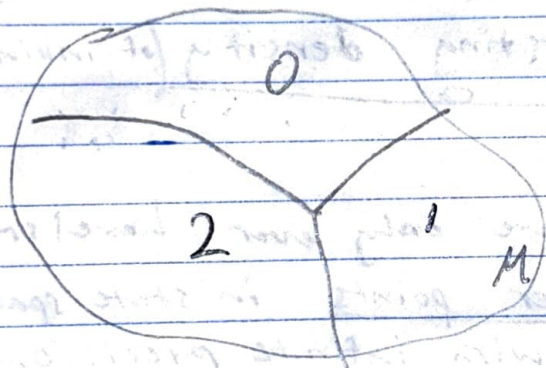
→ We now need to define an auxiliary characteristic function

→ For our cloud of points around i in state space M



Simply put, coarse graining is subdividing state space into regions
 ↳ however, we don't pick an arbitrary coarse graining; we rather use the naturally - found coarse graining
 → use stable and unstable manifold that are natural subdivisions of state space

Characteristic function
$$X_i = \begin{cases} 1 & x \in M_i \\ 0 & x \notin M_i \end{cases}$$



$$M = \{M_1 \cup M_2 \cup M_3\}$$

$$\mu_i = \int_M dx X_i(x) p(x) = \int_{M_i} dx p(x)$$

↑ over whole space using characteristic function
 → partitions are given by the dynamical systems

$$\sum_i \mu_i = \mathbb{I} \quad \Delta \mu_i = \int_M dx X_i(x) p(x) = \int_{M_i} dx p(x) = \Delta \mu_i p(x)$$

→ The partitioning & coarse graining procedure converts a space of infinite states into a finite number of regions / characteristics

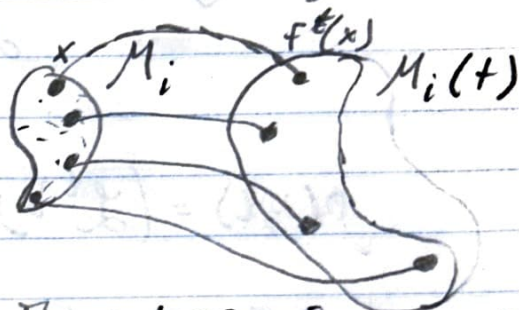
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Perron - Frobenius Operator

→ coarse graining is essentially discretizing our density function into an array of numbers

$$p(x) \rightarrow \begin{pmatrix} \Delta\mu_1 \\ \Delta\mu_2 \\ \vdots \\ \Delta\mu_N \end{pmatrix}$$

→ vector of all probabilities of the particle from the cloud of being in that coarse-grained region



→ let $f^t(x)$ represent a finite time map of trajectories through phase space

→ let J^t be the Jacobian at time t

The volume of our neighborhood is deformed through time

↙ conservation of mass in phase space $\Leftrightarrow$ determinism

$$\int_{f^t(M_i)} dx p(x, t) = \int_{M_i} dx_0 p(x, t=0)$$

but we want this integral in a single

Consequence of determinism in our system every traj. in state space continues to exist for all time t ,

$$\int dx = \int dx_0 |\det J^t|$$

$$J_{ij}^t = \frac{\partial x_j(t)}{\partial x_i(0)}$$

Set of coordinates
↓
Use Jacobian

$$p(x, t) = \frac{p(x_0, 0)}{\|\det J^t(x_0)\|} \quad (\star)$$

→ Symplectic / Liouville system $\Rightarrow \|\det J^t\| = 1 \forall t$

→ attracting region $\Rightarrow \|\det J^t(x_0)\| < 1$, density increases as trajectories accumulate

→ repelling region $\Rightarrow \|\det J^t(x_0)\| > 1$, density falls

→ Now, we define the operator

$$p(x, t) = [\mathcal{L}^t \circ p](x) : p(x, t) = \frac{p(x_0, 0)}{\|\det J^t(x_0)\|}$$

$$= \int dx_0 \delta(x - f^t(x_0, 0)) p(x_0, 0) \quad (*)$$

dirac delta $\Leftrightarrow$ all points / trajectories are unique

$$p(x, t) = (\mathcal{L}^t \circ p)(x) = \int dx_0 \delta(x - f^t(x_0, 0)) p(x_0, 0)$$

→ Recall the dirac delta integral:

$$\int_{M_i} dx \delta(x) = 1 ; \int_{M_i} dx \delta(h(x)) \quad \text{let } h(x^*) = 0$$

$$\int_{M_i} dx \frac{\delta(x)}{|h'(x^*)|} = \frac{1}{|h'(x^*)|}$$

$\Rightarrow$ Since we reproduce

(*) we have verified the expression (*)

in higher dims $\hookrightarrow$ this becomes $|\det J|!$

$$\int dx_0 \delta(x - f^t(x_0, 0)) p(x_0, 0)$$

Frobenius-Perron Operator

(generalization of a Fourier transform)

$$\Rightarrow p(y, t) = \int dx \mathcal{L}^t(y, x) p_0(x)$$

→ Linear Operator

Classical / deterministic mechanics: $\mathcal{L} \rightarrow \delta$

Quantum: $\mathcal{L} \rightarrow U^\dagger$ unitary operator

Stochastic: $\mathcal{L} \rightarrow$ Gaussian / spreading function